

Section A: Structured Questions [80 marks]

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Question 1 [10 marks]

(a) Figure 1.1 shows part of the cell body of a neurone.

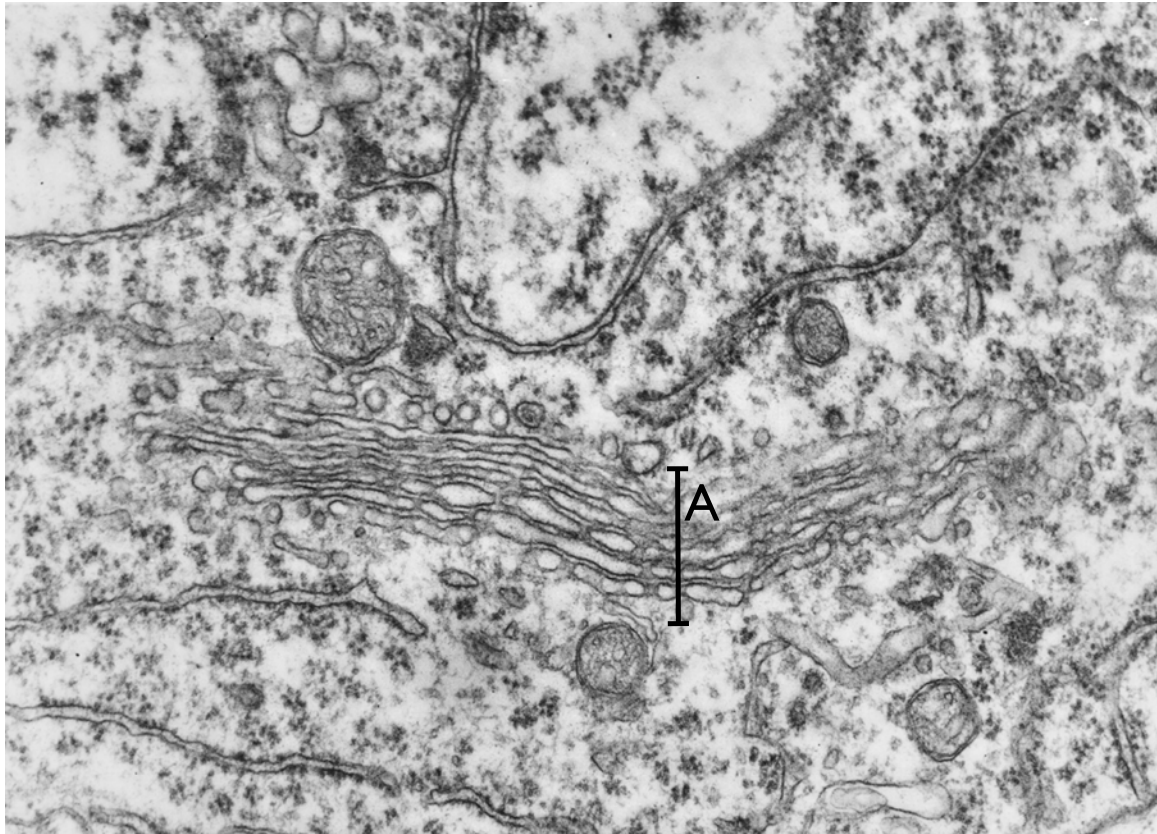


Figure 1.1

(i) Identify the organelle labeled A. [1]

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(ii) Relate the importance of this organelle to the cell. [2]

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The following structure in Figure 1.2 is also found in the axon of a neurone.

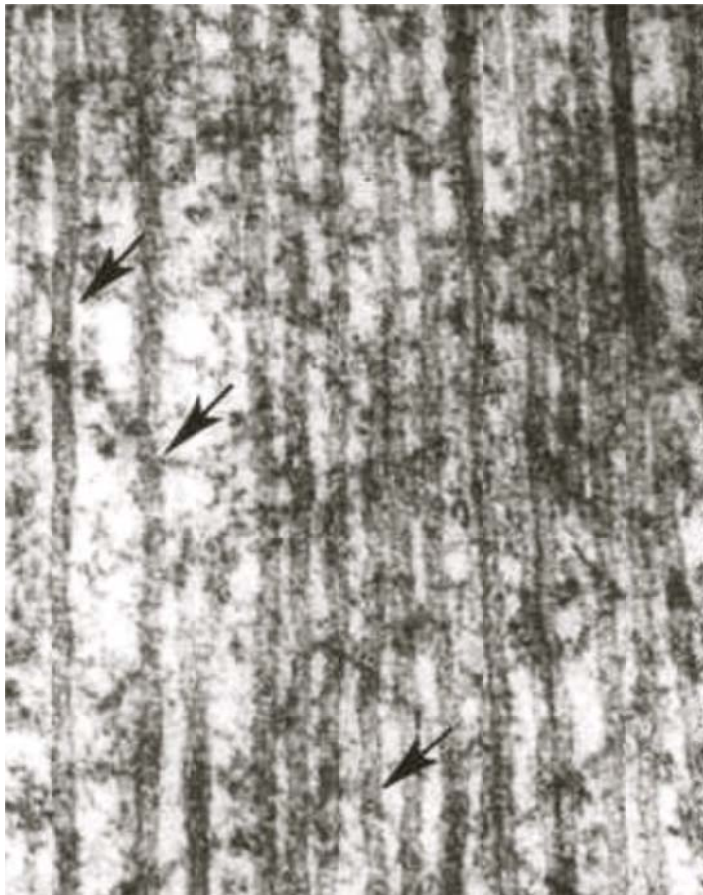


Figure 1.2

(iii) Identify the structures as indicated by the arrows in the diagram. [1]

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(iv) In what way is this structure related to the function of organelle A in the neurone? [1]

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(b) Figure 1.3 shows a cholinergic synapse of the neurone.

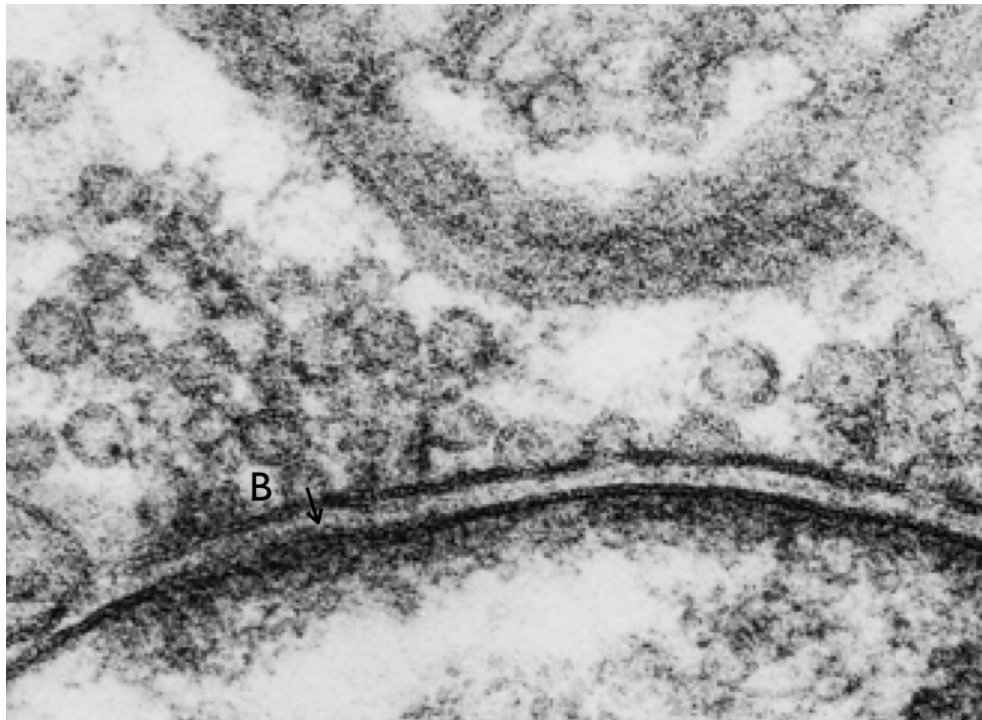


Figure 1.3

(i) Identify process B. [1]

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(ii) With your understanding of the cell membrane, explain how process B occurs. [1]

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(iii) Suggest the significance of this process with regard to synaptic transmission. [1]

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(iv) On the pre-synaptic membrane, there are also Ca^{2+} voltage-gated channels found. Describe how a change in potential difference will result in the diffusion of Ca^{2+} . [2]

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Question 2 [13 marks]

(a) Figure 2.1 below shows the transcription initiation process.

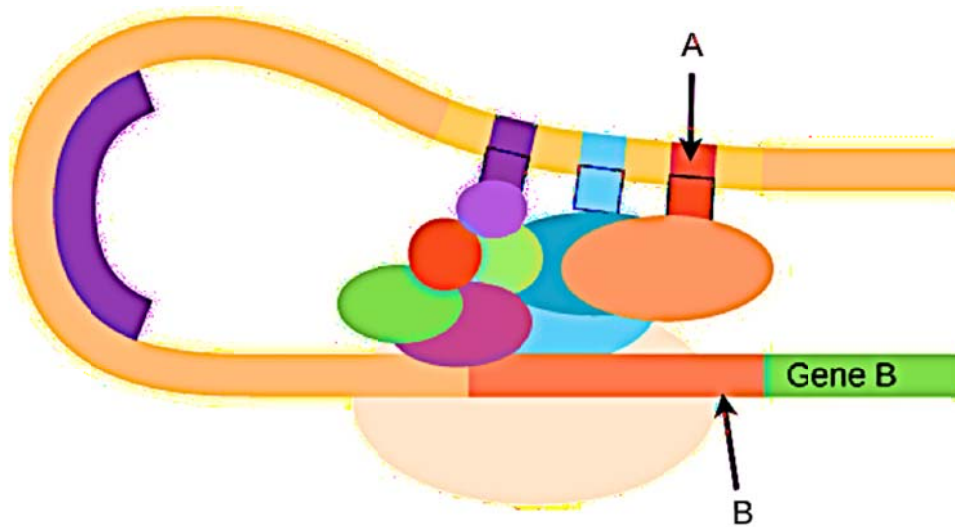


Figure 2.1

(i) Identify structures A and B [1]

A

B

(ii) With reference to Figure 2.1, describe how transcription is initiated and regulated. [3]

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(b) Figure 2.2 is an electron micrograph depicting the process of translation in a prokaryote.

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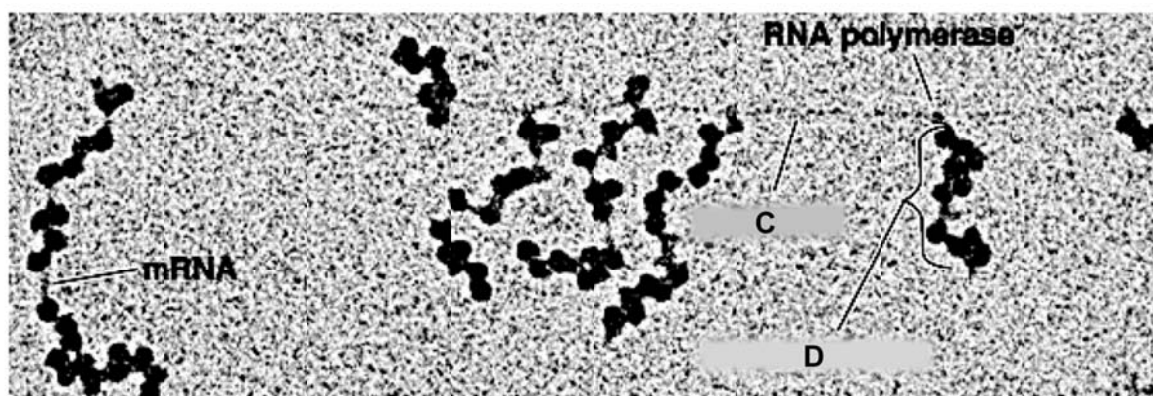


Figure 2.2

(i) Identify structures C and D. [1]

C

D

(ii) Describe the importance of structure D in translation. [1]

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(iii) With reference to Figure 2.2, suggest a difference between the translation in prokaryotes and eukaryotes. [1]

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(c) Different genetic mutations on a gene and its mRNA may result in different diseases. For example, RNA splice site mutation of β -globin mRNA is responsible of some cases of β -thalassemia and a point mutation on β -globin gene is responsible for sickle cell anaemia.

(i) Explain the significance of the point mutation to the properties of haemoglobin. [3]

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- (ii) Explain why a point mutation on a β -globin gene does not always produce mutated globin whereas an RNA splice site mutation will always produce mutated globin. [3]

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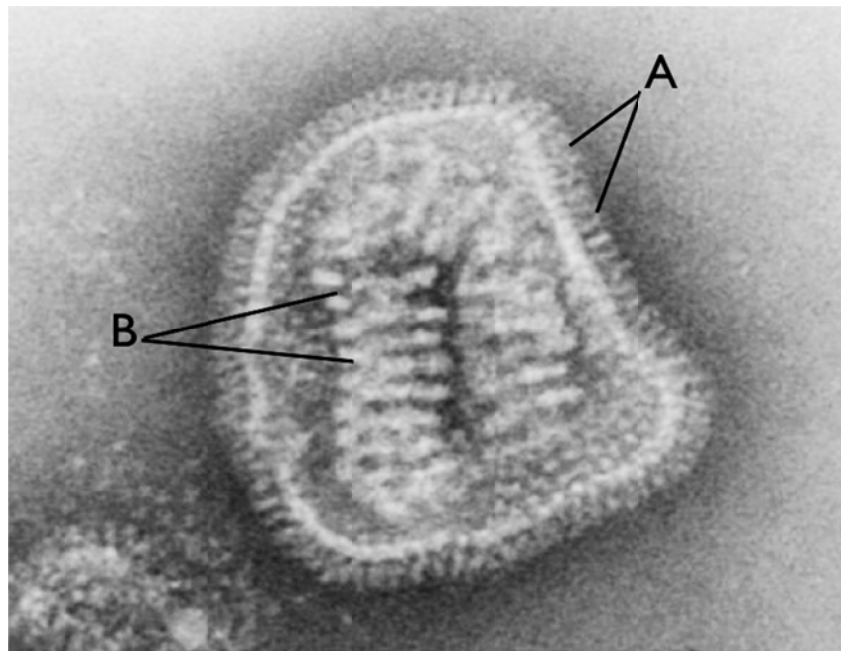
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Question 3 [13 marks]

Figure 3.1 shows an electron micrograph of H1N1.

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Figure 3.1

(a) Identify the components labeled A and B. [3]

A

B

(b) RNA-dependent RNA polymerase can be found in this virion while reverse transcriptase can be found in retrovirus. Besides the type of virus, describe one similarity and two differences between RNA-dependent RNA polymerase and reverse transcriptase. [3]

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Oseltamivir, better known under its trademark name as Tamiflu, is a recommended medication for H1N1. Its mechanism of action is illustrated in Figure 3.2 and 3.3.

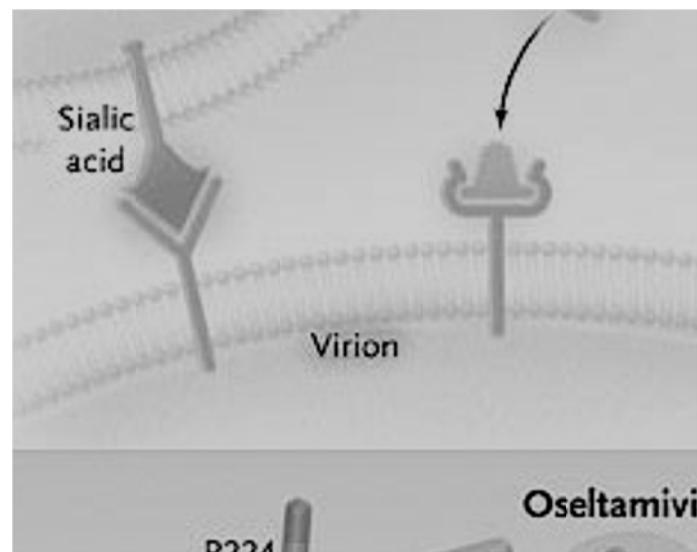


Figure 3.2

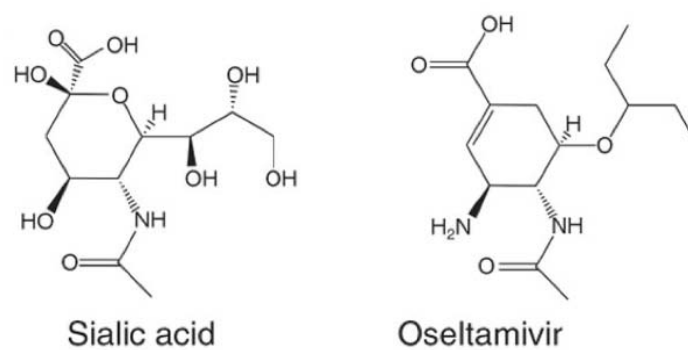


Figure 3.3

(c) Explain how oseltamivir affects the reproduction cycle of H1N1. [2]

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The effects of oseltamivir were studied on patients infected with influenza virus. These patients were randomly separated into two groups – one group administered with a placebo (a pseudo-drug with no effect) while the other group was administered oseltamivir. Symptoms were measured and recorded in Table 3.4.

Symptoms	Study Groups	
	Placebo (n=129)	Oseltamivir, 75mg (n=124)
Cough		
Duration, hr	55 (36-73)	31 (24-42)
Severity, arbitrary units	110	67
Nasal Obstruction		
Duration, hr	43 (31-64)	33 (25-43)
Severity, arbitrary units	80	60
Sore Throat		
Duration, hr	21 (8-29)	10 (4-19)
Severity, arbitrary units	29	20
Fatigue		
Duration, hr	41 (26-50)	24 (19-34)
Severity, arbitrary units	89	54
Feverishness		
Duration, hr	23 (16-29)	10 (8-14)
Severity, arbitrary units	49	21

Table 3.4

Table abstracted from: <http://jama.jamanetwork.com/article.aspx?articleid=192425>,
Efficacy and Safety of the Oral Neuraminidase Inhibitor Oseltamivir in Treating Acute Influenza - A Randomized Controlled Trial

(d) With reference Table 3.4, determine the effectiveness of oseltamivir. [2]

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(e) In a separate case, a patient was treated for H1N1. During this treatment, he was infected with H3N3. Upon testing, it was found that his blood now contains H3N3 and a new strain of virus, H1N3. Explain how this new strain could have arisen. [3]

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Question 4 [10 marks]

- (a) In maize plants, a recessive mutant allele leads to green leaves as opposed to bronze leaves. Another recessive mutant allele at a separate locus leads to yellow cobs as opposed to brown cobs. F1 maize plants from a standard dihybrid cross was test crossed with a plant with green leaves and yellow cobs. The table shows the phenotypes of 200 of the offspring.

Bronze leaves and yellow cobs	88
Bronze leaves and brown cobs	12
Green leaves and yellow cobs	14
Green leaves and brown cobs	86

- (i) In the space below, draw a genetic diagram to explain the observed results of the cross. [4]

(ii) Explain the results of the test cross. [3]

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(b) A chi-squared test was also carried out for the above cross and the chi-square value was 109.6. Using the probability table given, does this value for the chi-squared test support the expected phenotypic ratio? Explain. [3]

Degrees of freedom	Probability, p				
	0.10	0.05	0.02	0.01	0.001
1	2.71	3.84	5.41	6.64	10.83
2	4.61	5.99	7.82	9.21	13.82
3	6.25	7.82	9.84	11.35	16.27
4	7.78	9.49	11.67	13.28	18.47

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Question 5 [13 marks]

(a) Figure 5.1 below shows the blood glucose concentration measured.

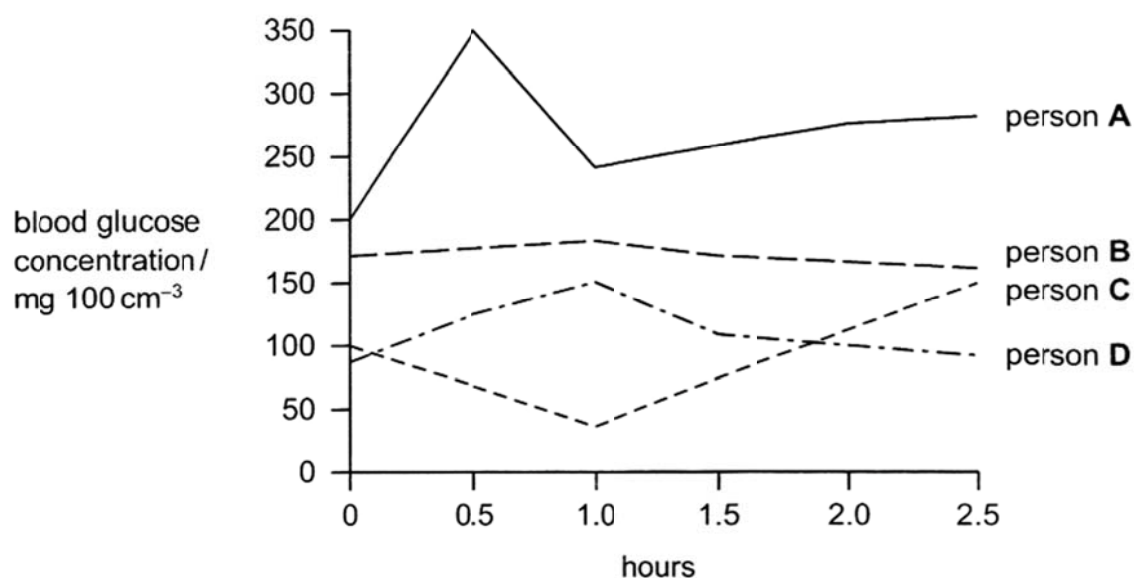


Figure 5.1

- (i) Identify which individual has a normal homeostatic system in controlling blood glucose level. Explain your answer. [2]

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- (ii) Describe and explain the condition of person C in the period between 0 – 1.0 hours. [2]

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- (iii) State the type of receptor that glucagon binds to and explain how this receptor is the fully activated. [2]

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- (b) Ras is a GTP-binding protein that helps to broadcast signals from the cell surface to other parts of the cell to stimulate cell proliferation. If Ras function is inhibited, the cell proliferation responses normally induced by the activated receptors do not occur. Conversely, if a hyperactive mutant, Ras protein, is introduced into some cell lines, the effect on cell proliferation is the same as that induced by the binding of ligands to cell surface receptors. This hyperactive product of mutant ras gene promotes the development of cancer.

Figure 5.2 below shows the sequence of events in a Ras cell cycle-stimulating pathway.

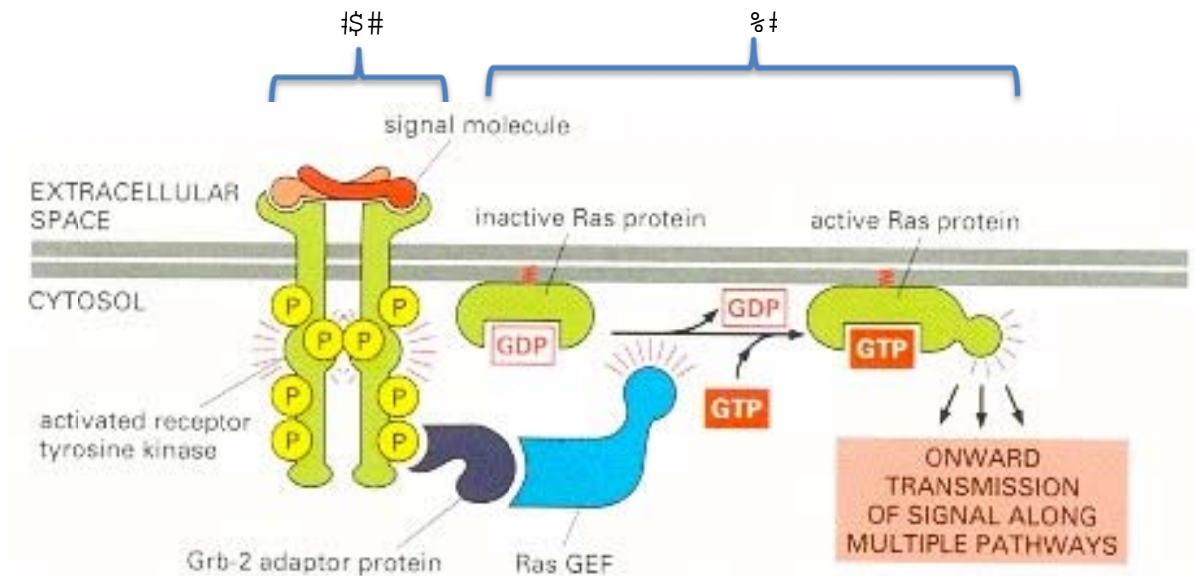


Figure 5.2

- (i) Briefly describe the stages A and B shown in Figure 5.2. [4]

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- (ii) Ras protein is a plasma membrane bound G-protein. Suggest how the activated G-protein can relay the signal from the ligand to activate cellular responses. [2]

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(c) Suggest one function of signal transduction. [1]

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Question 6 [10 marks]

Figure 6.1 shows an electromicrograph of a chloroplast

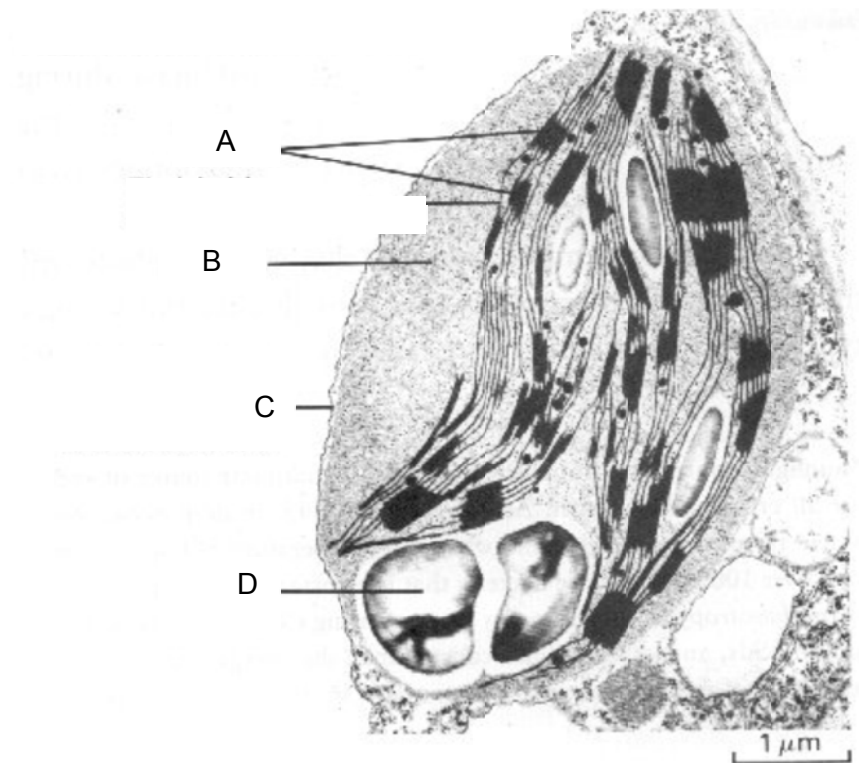


Figure 6.1

(a) Identify the features: [2]

A B

C D

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- (b) The blue dye DCPIP can be converted to colourless reduced DCPIP by gaining electrons.

A suspension of chloroplasts was made by grinding fresh leaves in buffer solution and centrifuging the mixture. Tubes were then prepared and treated in the following way:

Tubes	Content	Treatment	Colour	
			Start	After 20 min
A	1 cm ³ chloroplast suspension 5cm ³ DCPIP	Illuminated strongly	Blue-green	Green
B	1 cm ³ buffer solution 5cm ³ DCPIP	Illuminated strongly	Blue	blue
C	1 cm ³ chloroplast suspension 5cm ³ DCPIP	Left in the dark	Blue-green	Blue-green

- (i) Explain this observation using your knowledge on light-dependent reactions of photosynthesis. [4]

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- (ii) Explain, using tubes A and C, the effects of illumination on Calvin cycle (dark reaction). [2]

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- (iii) The chloroplast suspension may be contaminated with mitochondria. Explain the evidence from the investigation that presence of mitochondria was not responsible for the reduction of DCPIP. [2]

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Question 7 [11 marks]

The Galápagos tortoise or Galápagos giant tortoise (*Chelonoidis nigra*) is the largest living species of tortoise and 13th-heaviest living reptile, reaching weights of over 400 kg and lengths of over 1.8 metres. With life spans in the wild of over 100 years, it is one of the longest-lived vertebrates.

The tortoise is native to seven of the Galápagos Islands, a volcanic archipelago about 1,000 km west of the Ecuadorian mainland.

Shell size and shape vary between populations. On islands with humid highlands, the tortoises are larger, with domed shells and short necks. Due to the climate, there is abundant vegetation and vegetation is closer to the ground. On islands with dry lowlands, the tortoises are smaller, with "saddleback" shells, which are flatter than domed shells, and long necks. There is less vegetation. Charles Darwin's observations of these differences on the second voyage of the Beagle in 1835, contributed to the development of his theory of evolution.

Tortoise numbers declined from over 250,000 in the 16th century to a low of around 3,000 in the 1970s. This decline was caused by exploitation of the species for meat and oil, habitat clearance for agriculture, and introduction of non-native animals to the islands, such as rats, goats, and pigs.

- (a) (i) Explain how the distinct phenotypic differences between the populations may have arisen. [4]

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- (ii) State the phenomenon that describes the decline in tortoise numbers from the 16th century to the 1970s. [1]

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(iii) Suggest a disadvantage to a reduction in population size. [1]

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Figure 7.1 below shows the relationship between number of substitutions in cytochrome c between related pairs of species and how long ago the two compared species had diverged from a common ancestor.

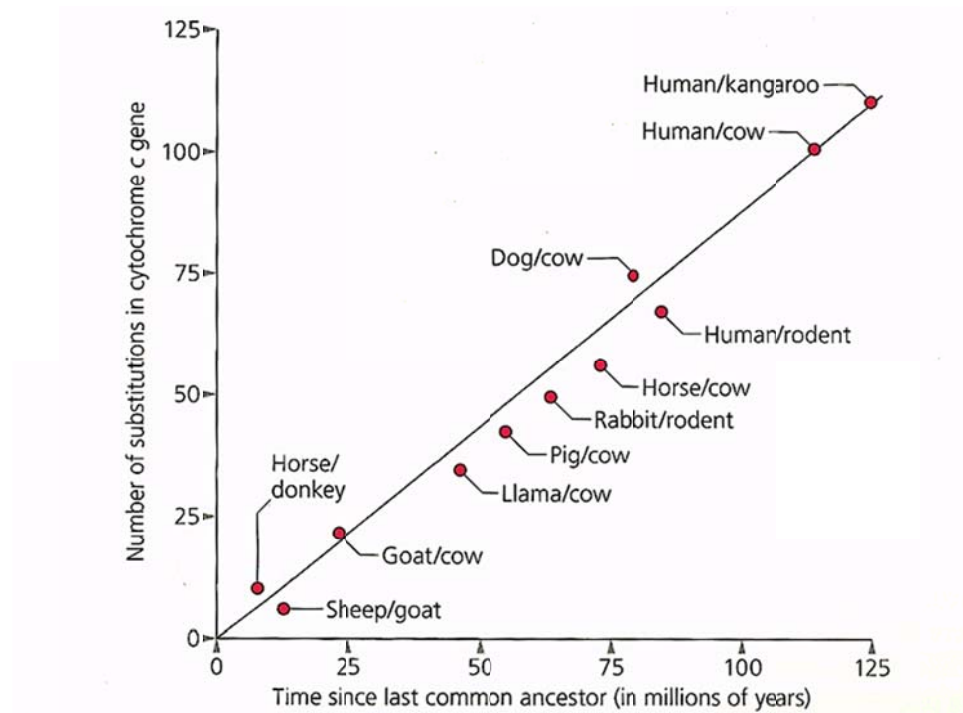


Figure 7.1

(b) Describe and explain the conclusions that can be drawn from Figure 7.1. [3]

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(c) Suggest the difficulties that arise when using the molecular clock. [2]

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Section B: [20 marks]

Answer **only ONE** question.

Write your answers on the separate writing paper provided. Your answers should be illustrated by large, clear labeled diagrams, where appropriate. Your answers must be in continuous prose, where appropriate.

Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

Question 8

- (a) Explain how the structure of triglyceride is related to its function. [7]
- (b) Compare and contrast between collagen and cellulose. [7]
- (c) Some enzymes function optimally in acidic conditions, while other enzymes function optimally in alkaline conditions. Explain why different enzymes have different optimal pH. [6]

Question 9

- (a) Explain the role of the F plasmids in bacterial conjugation and suggest the advantages of conjugation over transduction. [8]
- (b) Explain how the lac operon is regulated by both negative and positive control. [8]
- (c) Suggest why operons are necessary in bacteria. [4]